

02

Defect Types & Characteristics

Comprehensive Study Notes

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Chapter 2: Defect Types & Characteristics

Classification of Defects:

- Inherent Defects during Solidification from Liquid to Solid.
- Induced Defects during primary processing.
- Induced defects during secondary processing.
- Defects in service.

Primary Processing

- Metals and Alloys

Defects can occur during the primary processing of metals and alloys due to a variety of factors:

1. **Inclusions:** These are non-metallic particles or impurities that are present in the metal or alloy. They can be introduced during the manufacturing process or come from the raw materials. Inclusions can lead to weaknesses in the material and reduce its mechanical properties.
 2. **Porosity:** This is the presence of voids or gas pockets within the metal or alloy. Porosity can occur due to inadequate pouring or casting practices, or because of excessive gas content in the melt. Porosity can weaken the material and reduce its ductility.
 3. **Cracks:** These are fractures or fissures in the material that can occur due to improper cooling, excessive deformation, or other factors. Cracks can reduce the strength of the material and make it more susceptible to further damage.
 4. **Segregation:** This is the uneven distribution of alloying elements or impurities within the material. Segregation can occur due to inadequate mixing or because of differences in the melting temperature of different components in the alloy. Segregation can lead to non-uniform mechanical properties across the material.
 5. **Surface defects:** These are defects that occur on the surface of the material, such as scratches, pits, or other imperfections. Surface defects can be caused by improper handling or storage, inadequate cleaning, or processing problems.
- Ceramics

Defects can occur during the primary processing of ceramics due to various reasons:

1. **Inhomogeneity:** This is the uneven distribution of components or impurities within the ceramic material. Inhomogeneity can occur due to inadequate mixing or because of differences in the melting temperature of different components in the ceramic.
2. **Cracks:** These are fractures or fissures in the ceramic that can occur due to thermal gradients during processing, thermal shock, or excessive stresses. Cracks can reduce the strength of the material and make it more susceptible to further damage.
3. **Warping:** This is a deformation of the ceramic material that results in a change in its shape or dimensions. Warping can occur due to differences in thermal expansion rates

between the different components of the ceramic material or inadequate processing conditions.

4. **Delamination:** This is the separation of layers within the ceramic material. Delamination can occur due to poor adhesion between the layers or inadequate processing conditions.
5. **Porosity:** This is the presence of voids or pores within the ceramic material. Porosity can occur due to inadequate compaction or processing conditions, or because of excessive gas content in the material.
6. **Contamination:** This is the introduction of impurities or foreign particles into the ceramic material. Contamination can occur due to inadequate processing conditions or because of impurities in the starting materials.

- **Polymers**

Defects can occur during the primary processing of polymers due to various reasons:

1. **Inclusions:** These are foreign particles or impurities that are present in the polymer. They can be introduced during the manufacturing process or come from the raw materials. Inclusions can lead to weaknesses in the material and reduce its mechanical properties.
2. **Voids:** These are empty spaces or air pockets within the polymer. Voids can occur due to inadequate processing conditions, improper mixing or compounding, or because of the presence of gases in the polymer.
3. **Flow marks:** These are surface irregularities or marks on the polymer surface that occur due to uneven flow during processing. Flow marks can be caused by inadequate mold design, processing conditions, or cooling rates.
4. **Surface defects:** These are defects that occur on the surface of the polymer, such as scratches, pits, or other imperfections. Surface defects can be caused by improper handling or storage, inadequate cleaning, or processing problems.
5. **Delamination:** This is the separation of layers within the polymer material. Delamination can occur due to poor adhesion between the layers or inadequate processing conditions.
6. **Contamination:** This is the introduction of impurities or foreign particles into the polymer material. Contamination can occur due to inadequate processing conditions or because of impurities in the starting materials.

Secondary Processing

- Joining (Welding)
- Machining
- Heat Treatment
- Coating
- Cladding

Service Defects

A failure occurs when a component is unable to carry out the intended function satisfactorily due to fracture, deformation or deterioration.

- Design deficiencies
- Materials imperfections arising from manufacturing or fabrication.
- Processing deficiencies
- Assembly errors
- Improper conditions due to environmental and service conditions

Manufacturing Defects

EX/ Wrought VS Cast Alloys

- Wrought alloys are non-ferrous metal alloys with a composition that makes them extremely ductile. Unlike cast alloys, they are therefore particularly suitable for hot and cold forming processes such as drawing, rolling, and pressing. To produce wrought alloys, the non-ferrous metals first undergo continuous casting or ingot casting to form a preliminary shape, which is then further processed by hot or cold forming.
- Cast Alloys are those which can't be metal formed due to the lack in ductility necessary for extensive deformation. Formed by casting or semi centrifugal casting. Cast alloys usually has relatively low melting temperatures such as those with compositions around the Eutectic reactions in binary and ternary alloy systems. They have relatively high fluidity. Cast Alloys are only subjected to subsequent machining, trimming, grinding, etc.

Manufacturing Operation	Process/Product	Possible Defects
Foundry	Cast Ingot/Billet, Sand, Die, Investment Casting	Pipe, Blowhole, Gas Porosity, Shrinkage Porosity, Shrinkage Cavity/Crack, Non-Metallic Inclusion, Oxide/Dross Inclusion, Segregation, Internal Fissure, Ingotism, Scabs, Ingot Crack, Hot Tear, Cold Shut, Misrun
Wrought Processing	Rolled Bars/Rods/Plates/Sheets	Seam, crack, oxide discontinuity, non-metallic inclusion discontinuity, delamination, orange peel, exfoliation, transverse fine crack, edge crack
	Forged Bars/Rods	Forging burst, lap /fold, cold shut, push through defect, crack, rupture, tear cooling crack, thermal crack, asymmetrical /undesirable grain flow pattern, orange peel/alligator's skin, surface undulation, surface grain coarsening & discoloration (in Al alloys), hair line crack, decarburization, overheating, burning, cracking due to segregation, low mechanical strength properties due to improper grain flow/insufficient section reduction, blistering, rough surface, central burst
	Extruded Sections/Profiles/Tubes	Heat checks, radial and circumferential cracking, central burst, piping, delamination, peripheral

		coarse grain envelope, longitudinal discontinuity, annular defect, hot and cold shortness, cracking due to inclusion/segregation and gas voids, blistering due to entrapped gas in extruded bar stock/improper lubrication, funnel formation
	Drawing	Internal burst, transverse cracking, poor surface finish, wrinkling, orange peel, stretch-strain, edge cracking, season cracking (in brass).
	Swaging	Overlapping dimple, orange peel, surface tearing, micro fissure/crack, wrinkling, fold.
Assembly	Welding	Incomplete penetration, incomplete fusion, crack, undercut, slag inclusion oxide film, void, porosity, shrinkage, coarse grained/widmanstatten structure, nugget cracking, cavitation, intergranular crack (weld decay), distortion, improper weld profile, warpage, excessive concavity.
	Machining	Mechanical defects such as nick, scratch, ridge, etc.

		Grinding crack, burn/discoloration, hot spots, rehardening effect, softening effect, residual stresses, loss of dimension and parallelism, surface irregularities, distortion, undercutting and rough surface (in chemical milling), hydrogen embrittlement/intergranular attack or pitting and corrosion (in electrochemical machining), craters/voids/rough surface, hard chilled white layer, crack, heat affected zone (in electro discharge machining).
	Heat Treatment	Quenching crack, overheating, coarse grained/widmanstatten structure, burning, decarburization, oxidation /scale, hairline crack, retained austenite, soft spots, untampered martensite, distortion, warpage, low strength properties (in steels), crack, high temperature oxidation, discoloration, surface blisters, diffusion of cladding, distortion, overheating, low strength properties, orange peel effect (in Al alloys).
	Plating	Imperfect adhesion, blisters, flaking of coating, discoloration, crack,

		embrittlement, rough deposits (dark or granular).
	Pickling	Embrittlement cracks, pits.
	Forming	Crack, wrinkling, distortion.
Service	All Products	Fatigue crack, corrosion pits, intergranular corrosion cracking, corrosion fatigue, fretting corrosion, stress corrosion cracking, hydrogen embrittlement, exfoliation, galvanic corrosion, crevice corrosion thermal fatigue, hot corrosion fatigue, creep, brittle fracture.

Casting Defects

1. Porosity

- Gas porosity or blow holes are caused by accumulated gas or air which is trapped by the metal.
- These discontinuities are usually smooth-walled rounded cavities of a spherical, elongated, or flattened shape.
- If the sprue is not high enough to provide the necessary heat transfer needed to force the gas or out of the mold, the gas or air will be trapped as the molten metal begins to solidify.
- Blows can also be caused by sand that is too fine, too wet, or by sand that has a low permeability so that gas cannot escape.

2. Segregation

- Segregation means departure from the average composition.
- Macro segregation refers to the uneven distribution of elements or phases in the casting on a relatively large scale, typically visible to the naked eye. It occurs when heavier elements, such as iron or copper, settle to the bottom of the mold during solidification, creating regions of higher concentration in some areas of the casting and lower concentration in others.
- Micro segregation, on the other hand, refers to the uneven distribution of elements or phases in the casting on a smaller scale, typically not visible to the naked eye but can be detected under a microscope. It occurs when there are variations in the solidification rate of different parts of the casting, causing solute-rich regions to form in some areas and solute-poor regions to form in others.
- Positive Segregation (+): if the concentration is greater than average.

- Negative Segregation (-): if the concentration is less than average,
 - Cause for Segregation is differential solidification of liquid solution.
 - Example: Steel contains C, S, P, Si and Mn; so, it is prone to segregation during solidification.
3. **Sand Inclusions and Dross**
- They are nonmetallic oxides, which appear on the radiograph as irregular dark blotches.
 - These come from disintegrated portions of mold or core walls and/or from oxides (formed in the melt) which have not been skimmed off prior to the introduction of the metal into the mold gates.
4. **Cracks**
- Cracks are thin (straight or jagged) linearly disposed discontinuities that occur after the melt has solidified.
 - They generally appear singly and originate at casting surfaces.
5. **Shrinkage**
- Shrinkage is a form of discontinuity that appear as dark spots on the radiograph.
 - Shrinkage assumes various forms, but in all cases, it occurs because molten metal shrinks as it solidifies, in all portions of the final casting.
 - Shrinkage is avoided by making sure that the volume of the casting is adequately fed by risers which sacrificially retain the shrinkage.
 - There are four types of shrinkage:
 - Cavity: Cavity shrinkage defects occur when the metal cools and solidifies before filling the entire mold cavity, resulting in a void or cavity within the casting. This type of defect is typically caused by insufficient molten metal being poured into the mold, poor gating design, or a lack of venting.
 - Dendritic: Dendritic shrinkage defects occur when the metal solidifies and contracts, causing dendrites (finger-like projections) to form in the solidifying metal. These dendrites can intersect and create voids within the casting, resulting in weak spots or cracks. This type of defect is often caused by high cooling rates, which can lead to rapid solidification and dendrite formation.
 - Filamentary: Filamentary shrinkage defects occur when thin, elongated channels or voids form within the casting. These defects can be caused by insufficient feeding of the mold, high cooling rates, or a lack of sufficient risers to provide additional molten metal during solidification.
 - Sponge: Sponge shrinkage defects occur when large voids or pores form within the casting, giving it a spongy appearance. This type of defect is often caused by gas bubbles trapped within the molten metal during pouring, which expand as the metal cools and solidifies.

6. Wash and Cuts

- Wash: A "wash" is a type of casting defect that occurs when a thin layer of mold material erodes or wears away due to the action of molten metal during pouring. This results in a thin layer of metal adhering to the mold surface and creating a rough, pitted surface on the casting. Causes of wash defects include high pouring temperatures, high turbulence of the molten metal, high velocity of the molten metal, or insufficient coating or refractory material.
- Cuts: A "cut" is a type of casting defect that occurs when the mold material is damaged or removed during the casting process, creating a groove or channel in the casting. Cuts can be caused by a variety of factors, including inadequate mold design or construction, improper gating and rising, excessive mold movement during pouring or solidification, or mechanical damage to the mold. Cuts can also occur due to the presence of impurities in the metal or gas porosity, which can create weak spots in the casting.

7. Hot Tears

- Occur during cooling of the solid cast due to induced internal stresses caused by dimensional changes (shrinkage).
- It occurs in alloys with extended freezing range.
- When solid and liquid coexist over a wide range of temperature, contraction of the solid around the rigid mold surface can initiate rupture in partially solidified regions.
- Internal stresses arising from hindered contraction due to core or mold resistance.
- Internal stresses arising from differential cooling due to improper casting design.
- Prevention:
 - reducing pouring temperatures
 - Reducing the die temperature
 - Increasing pressurization time
 - Increasing draft angles on the casting

Cast Iron (1.8 – 4.5 % C)

Cast Alloy:

- Excellent wear Resistance
- Si is added to the alloy 0.5-3.5%
- Si < 1%: improves fluidity.
- Si > 1-3.5% improves fluidity + stabilizes Graphite.

1. White Cast Iron (Ferrite + Fe₃C)

White cast iron is characterized by a microstructure of primary cementite (Fe₃C) and pearlite. The high carbon content of white cast iron makes it very hard and brittle, with low ductility and impact resistance. It has high compressive strength but low tensile strength. This type of iron is mainly used for wear-resistant applications where high hardness is required, such as in the manufacture of grinding balls, rolls for crushing, and wear plates.

- Very hard, brittle, and difficult to machine due to the high-Volume fraction of Cementite phase.
- Mechanical properties for the unalloyed WCI
- TS: 350MPa
- Compressive strength: 2224, 2460 and 2220 MPa
- E= 175 GPa
- BHN: 375-600
- Used for nozzles, decorative furniture, ball mills, liners of cement mixers.

2. Malleable Cast Iron

Malleable cast iron is produced by heat-treating white cast iron, which transforms the hard, brittle microstructure into a ductile, tougher structure of graphite nodules in a ferrite or pearlite matrix. Malleable cast iron has high strength, good ductility, and excellent impact resistance. It is used for a wide range of applications, including automotive and agricultural parts, pipe fittings, and hand tools.

- Cast iron with Course carbon forming irregular round lumps known as temper CI.
- Obtained by Malleableization HT of White CI
- When Cu is Added to it, tensile strength and corrosion resistance increases.
- TS = 580 MPa
- E: 172 GPa
- Elongation % 2-15
- BHN= 135-155

- Railway components, frame suspension and wheels, steering components, and transmission of different parts

3. Grey Cast Iron (Ferrite + Graphite Flakes)

Gray cast iron is characterized by a microstructure of graphite flakes in a ferrite or pearlite matrix. The graphite flakes provide good lubrication properties and help to dampen vibrations, making gray cast iron ideal for use in applications such as engine blocks, brake drums, and machinery bases. Gray cast iron has low tensile strength but high compressive strength and good wear resistance.

- Easier in machinability
- Vibration damping (graphite Flakes)
- Notch insensitive
- TS: 250MPa
- BH: 179-202
- E: 105 Mpa
- Internal combustion engine housing, Valve bodies, electrical boxes, cookware

4. Nodular Cast Iron (Ferrite + Graphite Nodules)

Nodular cast iron, also known as ductile iron, has a microstructure of graphite nodules in a ferrite or pearlite matrix. The graphite nodules provide good strength and toughness, and nodular cast iron has high ductility and excellent impact resistance. It is used in a wide range of applications, including automotive and industrial parts, machinery components, and pipe fittings.

- Spherulitic CI, spheroidal CI
- Graphite Nodules (inhibits crack propagation)
- Higher strength than GCI
- Ductility up to 6% as cast and 20% in the annealed form.
- Low cost
- Simple and complex shapes
- Machinability is better than steel.

Primary Processing Defects

1. Rolling Defects

- Wavy Edges: Improper rolling speed
- Zipper Cracks-Central cracks: Too much rolling at center
- Seams-Edge Cracks: Over strain hardening, poor ductility, convex roll bending,
- Alligatoring
- Defects in the Original cast material or non-uniform deformation

2. Extrusion Defects

A compression forming process in which the work metal is forced to flow through a die opening to produce a desired cross-sectional shape.

- **Center Bursts:** internal cracks due to excessive tensile stress at the center (high die angle, low extrusion ratio). It occurs due to difference in velocity and tensile stresses in center of the material. Prevent by using appropriate die angle, friction, and extrusion ratio.
- **Piping:** sink hole at the end of the billet under direct extrusion
- **Surface Cracking:** High part temperature due to excessive extrusion speed and high strain rate. Surface cracks are developed due to high extrusion temperature, high ram speed, or high friction.

To prevent the extrusion defect of center bursts, a low extrusion ratio should be used. Center bursts occur when the center of the extruded material ruptures due to excessive tensile stress. This can happen when the material is being extruded with too high of an extrusion ratio, which is the ratio of the cross-sectional area of the billet to the cross-sectional area of the extruded product. When the extrusion ratio is too high, the material experiences a greater degree of deformation and elongation, resulting in increased tensile stresses at the center of the product. This can cause the material to rupture, leading to center bursts.

Non-uniform flow, also known as extrusion inhomogeneity, can occur in the extrusion process when there is an uneven distribution of material flow, resulting in variations in the cross-sectional dimensions of the extruded product. Prevent by proper selection of the billet material, optimal billet temperature, proper extrusion die design, proper lubrication, use of back-pressure.

Surface cracks (Chevron Cracks) and bursts developed in circular, rod extrusions and also in wire drawing.

Sink hole at the end of the billet occurs in hot extrusion (due to titled tail pipe or fish tail).

Heat check is a surface defect that can occur in the extrusion process. It appears as a series of cracks that are perpendicular to the extrusion direction and are typically located on the surface of the extruded product. Heat check is caused by thermal fatigue and is typically associated with high-temperature extrusion of certain materials such as aluminum alloys.

Peripheral Coarse Grain Problems

Large-grained recrystallized microstructure on the surface leads to:

- Issues in further processing
- Inferior Product Quality
- Significant machining hence waste material

A large-grained recrystallized microstructure on the surface of a material is typically due to high-temperature exposure, such as during heat treatment or hot working processes. During high-temperature exposure, the existing grains in the material can grow or recrystallize, leading to the formation of larger grains. Recrystallization occurs when the existing grains in the material undergo plastic deformation at high temperatures, resulting in the formation of new, strain-free grains. The recrystallized grains tend to be larger than the original grains because they are nucleated at the boundaries between the deformed grains. This can lead to a more coarse-grained microstructure on the surface of the material.

3. Forging Defects

- Surface Cracks
 - Extensive working at low temperature
 - High Sulfur content
- Crack at Flash
 - More prevalent for thinner flash
 - Penetrates to work piece.
- Internal Cracks
 - Internal stresses
- Cold Shuts or Folds
 - is a defect in forging in which small cracks appears at corner of forging. This defect is mainly caused due to improper design of the dies.
- Lubricant Residue
 - Entrapped on folding.
- Flakes
 - cannot be noticed from outside because these are internal defects. internal ruptures caused due to improper cooling of large forging. Flakes appears as bright silvery discontinuous in fracture surface.
- Scale Pits
- Die Shifts
- Residual Stresses
- Open die forging defects:
 - Fracture: Exhausted Ductility, Intergranular fracture in hot working.

- Barreling: Friction
- Solution: Limited deformation per step, process anneal between steps.

Secondary Processing Defects

1. Welding Defects

Defect	Possible Causes	Possible Solution
Incomplete Penetration	1. Faulty joint design 2. Welding speed too rapid. 3. Insufficient welding current or nozzle size 4. Too large an electrode/filler rod	Check root gap, root face.
Poor Appearance	1. Current too high or too low – flame too large or too small. 2. Incorrect use of electrode/blowpipe or filler rod. 3. Faulty electrode. Incorrect flame setting.	Check procedure and slope and tilt angles. Dry or change electrode. Readjust flame, clean nozzle.
Undercutting	1. Current too high/nozzle too large 2. Incorrect manipulation 3. Arc length too long 4. Welding speed too rapid	Check angles so that arc force is used to fill undercut.
Excessive Spatter	1. Current too high 2. Arc length too long 3. Arc blow 4. Faulty electrode	See arc blow.
Arc Blow	Magnetic field created when welding with direct current.	Use a.c. machine. Rearrange or split return clamp. Use copper back-up bar or magnet to oppose field. Change direction of welding. Change angle of electrode when deflection

		begins. Use small-gauge electrode and more runs.
Pin Holes	1. Contamination of joint 2. Damp electrodes	Remove paint, rust, scale etc. from work. Check electrodes for dampness.
Stag in Weld	1. Joint design contains too narrow an included angle. 2. High viscosity of molten metal. 3. Rapid chilling. 4. Too low a weld temperature.	Use pre-heat. Failure to remove a slag from previous weld in multirun welds.
Porous Welds	1. Weld speed too rapid. 2. Current too low. 3. High Sulphur or other impurities in metal. 4. Faulty electrodes.	Use low hydrogen electrodes or process.
Cracked Welds	1. Faulty electrodes. 2. Stressed welds. 3. Shape of weld bead incorrect. 4. Craters present. 5. Too fast a cooling rate.	1. Use low hydrogen electrodes. 2. Redesign work or jig or preheat. 3. Use slower travel speed. 4. Fill craters. 5. Post-heat as well as pre-heat.
Distortion & warpage	1. Incorrect design of weld 2. Overheating 3. Incorrect welding sequence	Preset work. Use skip or backup welding
Brittle Welds	1. Incorrect choice of electrode 2. Incorrect heat treatment 3. Air-hardening deposit 4. Base metal pick-up	

Heat Affected Zone

The heat affected zone (HAZ) is the region of the base material that is affected by the heat generated during welding. Some common characteristics of the HAZ in welding defects include:

- **Changes in microstructure:** The high temperatures generated during welding can cause changes in the microstructure of the base material in the HAZ. For example, the HAZ may undergo grain growth, phase transformations, or the formation of new phases, which can affect the mechanical properties of the material.
- **Residual stresses:** The thermal cycling and rapid cooling during welding can lead to the development of residual stresses in the HAZ, which can affect the deformation and fracture behavior of the material.
- **Hardness variation:** The HAZ can exhibit variations in hardness due to changes in microstructure and residual stresses. The hardness in the HAZ may be higher or lower than the hardness of the base material, depending on the specific welding process and parameters used.
- **Reduced toughness:** The HAZ can exhibit reduced toughness compared to the base material, which can make it more susceptible to cracking or failure under certain conditions.
- **Potential for cracking:** The HAZ is often a location of high stress concentration and can be prone to cracking, especially if the welding process is not optimized or the material is not properly prepared or cleaned before welding.

The heat affected zone (HAZ) in welding can be divided into several distinct sections, each with its own unique characteristics. These sections include:

- **Fully annealed zone:** This is the region of the HAZ that is farthest from the weld and is not affected by the heat generated during welding. The fully annealed zone has a fine, uniform grain structure and is typically the strongest and most ductile part of the material.
- **Recrystallized zone:** This is the region of the HAZ that has been heated above the recrystallization temperature but below the melting point. In this zone, the original grain structure is destroyed, and new, strain-free grains are formed. The recrystallized zone typically has a coarse, equiaxed grain structure and is generally weaker and less ductile than the fully annealed zone.
- **Partially annealed zone:** This is the region of the HAZ between the fully annealed and recrystallized zones. In this zone, some of the original grains are still present, but there are also some recrystallized grains. The partially annealed zone has a mixed microstructure and typically exhibits intermediate mechanical properties between the fully annealed and recrystallized zones.
- **Grain growth zone:** This is the region of the HAZ that has been heated above the recrystallization temperature and allowed to cool slowly. In this zone, the recrystallized grains grow larger due to the high temperature and slow cooling rate. The grain growth

zone typically has a coarse, elongated grain structure and is generally weaker and less ductile than the recrystallized zone.

2. Heat Treatment Defects

a) Weld Decay: Sensitization

EX/ Sensitization of Stainless Steel:

When Stainless Steels is heated to 400-900°C, Chromium (Cr) [Cr depletion region from grain body (anode)] forms carbides (Cr_{23}C_6) along the grain (cathode) boundaries at 650°C (chromium carbide precipitation), which results in depletion of Cr around and along grain boundaries, causing Intergranular Corrosion and failure.

- Stainless steel is a type of steel alloy that contains a minimum of 10.5% chromium (Cr). Chromium is a key element in stainless steel because it forms a passive oxide layer on the surface of the steel that protects it from corrosion. However, when stainless steel is exposed to high temperatures, such as during welding or heat treatment, it can become sensitized and lose its corrosion resistance.
- Sensitization of stainless steel occurs when chromium carbides (Cr_{23}C_6) form along the grain boundaries at temperatures between 400-900°C. This process is known as chromium carbide precipitation, and it occurs because carbon has a higher affinity for chromium than oxygen does. As the stainless steel is heated, carbon atoms diffuse from the steel and combine with chromium atoms to form chromium carbides.
- The formation of chromium carbides at the grain boundaries depletes the surrounding area of chromium, leaving it vulnerable to corrosion. This is because the chromium-depleted regions along the grain boundaries become anodic to the rest of the steel, leading to the initiation and propagation of corrosion. The loss of chromium from the grain boundaries reduces the ability of the stainless steel to form a protective oxide layer, leaving it susceptible to intergranular corrosion and failure.
- To prevent sensitization, stainless steel should be heated within the recommended temperature range and cooled rapidly to prevent the formation of chromium carbides. Additionally, the carbon content of the stainless steel should be minimized, and the chromium content should be maximized to ensure that there is enough chromium to form a protective oxide layer on the surface of the steel. Heat treatments such as solution annealing and quenching can be used to dissolve chromium carbides and restore the corrosion resistance of sensitized stainless steel.

b) Cracking During Heat Treatment

Cracking during heat treatment is a common processing defect that can occur in metals and alloys when they are subjected to thermal processing, such as annealing, quenching, or tempering. This defect is characterized by the formation of cracks or fissures in the material, which can lead to reduced mechanical strength.

EX/ Flash line cracking during heat-treatment of an Al-Cu alloy forged component, due to overheating.

- In the case of flash line cracking during heat treatment of an Al-Cu alloy forged component, this defect occurs due to overheating during the forging process, which can result in localized melting or deformation of the material. When the material is subsequently subjected to heat treatment, this can cause the formation of a brittle intermetallic compound at the flash line, which is a thin, sharp ridge that forms where the forging die meets the workpiece. This results in a weakened or failed component.
- Other potential causes of cracking during heat treatment include improper quenching, excessive thermal gradients, or inadequate preheating or cooling.

c) Incipient Melting during HT

- Incipient melting is a phenomenon that can occur during the heat treatment of materials, particularly metals and alloys. It refers to the partial melting of the material at the grain boundaries (creating ROSETTES), where the temperature is typically highest during the heat treatment process.
- Incipient melting of grain boundaries during heat treatment is a processing defect that can occur in metals and alloys when they are subjected to elevated temperatures for extended periods of time. This defect is characterized by the partial melting or softening of the grain boundaries within the material.
- This process occurs when the temperature of the material exceeds the melting point of the grain boundaries, which are regions of low atomic density that separate individual grains within the material. At these high temperatures, the grain boundaries can begin to melt or soften, allowing the individual grains to deform or shift position relative to each other.
- When incipient melting occurs, the grain boundaries can become weakened, resulting in a reduction in the mechanical strength and ductility of the material. This can lead to cracking or other forms of damage, particularly under stress. Incipient melting can also increase the susceptibility of the material to corrosion, as the weakened grain boundaries provide a pathway for corrosive agents to penetrate the material.
- The occurrence of incipient melting can be influenced by a range of factors, including the temperature and duration of the heat treatment process, the composition of the material, and the presence of impurities or other defects. The risk of incipient melting can be reduced through careful control of the heat treatment process, including the use of appropriate heating rates, holding times, and cooling rates.

d) **Sulfurization or Eutectic Phase Melting during HT**

Sulfurization or Eutectic Phase Melting is a type of heat treatment defect that occurs when sulfur is present in the alloy and reacts with the metal to form low-melting eutectic phases. This can occur when the alloy is exposed to high temperatures during heat treatment, leading to the formation of liquid phases at the grain boundaries. These liquid phases can then promote crack formation and propagation, leading to reduced mechanical properties and increased susceptibility to fracture.

e) **Quenched Cracks in Steel**

Quench cracking is a heat treatment defect that occurs when steel is cooled too rapidly during the quenching process. When steel is heated and then rapidly cooled, the outer layers of the steel cool and contract faster than the inner layers. This creates high tensile stresses on the surface of the steel, which can exceed the material's yield strength and result in cracking.

f) **Decarburization (White Surfaces)**

Decarburization is a common heat treatment defect that occurs on the surface of carbon and alloy steels. It is the process of loss of carbon from the surface layer of the steel, resulting in a white or grayish appearance. Decarburization occurs when the steel is heated in an oxidizing atmosphere such as air or a furnace with an inadequate protective atmosphere.

Decarburization can significantly reduce the surface hardness and strength of the steel, making it susceptible to failure or deformation. It can also affect the surface finish and machinability of the steel. Therefore, it is essential to control the furnace atmosphere and temperature during heat treatment processes to prevent or minimize decarburization. Protective atmospheres such as nitrogen or hydrogen can be used to minimize decarburization.

3. Machining Defects

a) Grinding Marks

Grinding Force + Excessive Force → Machining Defects will be introduced to a material with intrinsic defects (pores and impurities) → Creates crack propagation that will result in failure.

It is possible for grinding marks to contribute to the propagation of cracks in a material that already contains intrinsic defects, such as voids, pores, or impurities. When a material contains such defects, it is already weakened and more susceptible to fracture. Grinding marks can create stress risers on the surface of the material, which can act as initiation points for crack propagation. The cracks can then propagate through the material along the grain boundaries until they encounter a void, pore, or impurity, which can act as a stress concentrator and cause the material to fracture.

b) Surface Burning

Surface burns, craters, and rough surfaces are common machining defects that can occur during electrical discharge machining (EDM) processes.

- Surface burns occur when the material being machined is overheated during the EDM process, resulting in localized melting, or burning of the surface. This can be caused by a variety of factors, including excessive power or current settings, poor electrode wear, or improper flushing of the dielectric fluid. Surface burns can lead to reduced material strength and increased susceptibility to corrosion and other forms of degradation.
- Craters are small, bowl-shaped depressions that can form on the surface of the machined material. They are often caused by the repeated sparking of the EDM electrode at a specific location, which can result in localized material erosion. This can be caused by factors such as inadequate flushing of the dielectric fluid, poor electrode alignment or wear, or insufficient voltage or current settings. Craters can result in reduced surface finish and accuracy, as well as decreased mechanical strength and durability.
- Rough surfaces can result from a variety of factors, including improper electrode wear, inadequate flushing of the dielectric fluid, or excessive machining speed. This can result in irregularities and bumps on the surface of the machined material, which can reduce its functionality and aesthetics.

Defects	Effects
Design Deficiencies	
Wrong material/ heat treatment/joining technique.	Lowers mechanical properties including fatigue resistance, toughness, and wear, increases corrosion susceptibility.
Inadequate stress analysis of component or providing inadequate fastener/ stiffener/shimming.	Increases the chances of failure to withstand the service loads as well as susceptibility to fatigue.
Mechanical notches, sharp corners, sharp/abrupt section change or abnormal stress raisers.	Increases the stress concentration factor and reduces fatigue strength drastically.
Improper location of oil holes, vents, etc.	Introduces weakness for increased accumulation of stress intensity.
Improper assembly design/dissimilar material.	Increases load factors due to mismatch and imparts galvanic corrosion.
Unanticipated service condition.	Induces an element of overloading.
Improper casting-porosity, voids, shrinkage cavities, pipes, hot tear, cold shut, cracks, etc.	Causes unsoundness of the material, lowers the strength capacity to withstand the service load and can serve as a notch for a propagative type of crack.
Inclusion	Acts as a source for fatigue crack propagation and enhances corrosion susceptibilities.
Banding/Segregation	Renders the material discontinuous and susceptible to stress sensitivity by decreasing the fatigue resistance / toughness. Presence of micro and macro segregation causes the inherent problem of cracking either during heat-treatment or forging / rolling.
Improper forging-folds/laps, cold shut, thermal crack	Is highly dangerous as a state of pre-cracking which easily becomes a stress concentration factor for fatigue failure and susceptible to sudden overload failure by virtue of the reduced section thickness.
‘Push through’ defect	Acts as a discontinuity or crack in the normal grain flow pattern and is critical, particularly in the fillet radius and section thickness change area for easy development of fatigue crack.

Unfavorable grain flow/(kink/re-entrant) defect	Serves as a potential source of premature fatigue failure in service primarily due to pronounced anisotropy in strength properties and the intended strength not being realized in the location particularly end grains in the fillet areas can lead to initiation of fatigue cracks under fluctuating loading conditions in service.
Grain coarsening/grain boundary oxidation (or burning)	Renders material susceptible to brittleness in service. Parts bearing burning defect are totally wasted. Reduces the fatigue resistance property.
Decarburization/Scaling	Reduces surface fatigue properties drastically.
Improper rolling-seam, longitudinal discontinuity, crack, delamination, etc.	Acts as a mechanical notch and reduces fatigue strength property
Improper extrusion-heat checks, blisters, etc.	Serves as a mechanical notch for fatigue crack development, induces brittleness and acts as a potential notch creating a localized area of weakness.
Delamination	Results in the peeling effect.
Longitudinal Discontinuity	Acts as an oxide film discontinuity and reduces fatigue resistance property.
Coarse Grain Envelope	Causes cracking during forging of the extruded bar stack.
Processing Defects: Improper Machining	
Grinding Crack	Acts as a harmful defect/discontinuity whether shallow or deep, by way of stress raiser for development of premature fatigue failure.
Hot Spot	Constitutes localized surface weakness due to either the formation of brittle (over-heated) zones from the effect of high temperature or reduction in hardness due to decarburization. Such surface anomalies exert adverse effects on the fatigue as well as wear resistance property. particularly in case-hardened parts.

Improper Heat Treatment	
Overheating (coarse grained/widmanstatten structure in steel, blistering, eutectic melting/grain boundary cracking) (in aluminum alloys)	Renders structural deterioration induces brittleness and reduces fatigue strength as well as corrosion resistance property.
Burning	Renders product unacceptable for use.
Oxidation/Scaling	Reduces effective thickness and induces decarburization.
Quenching Crack	Reduces section thickness and fatigue strength significantly.
Under tempering	Increases susceptibility to brittle fracture.
Over tempering/Overaging	Softens the strength capacity and reduces fatigue strength.
High temperature oxidation (in Al alloys)	Causes structural weakness.
Decarburization	lowers the surface hardness and significantly reduces the endurance limit.
Excessive case thickness (case carburizing)	Develops micro crack in the transition zone between the case layer and the core.
Carbide network	Causes embrittlement and crack.
White layer (case nitriding)	Causes embrittlement and crack.
Retained Austenite	Softens the strength property as a localized weakness but develops brittleness/crack when converted into untampered martensite during tempering.
Improper Cleaning, Plating, and Coating	
Pitting, hydrogen embrittlement, galvanic corrosion, crack.	Deteriorates fatigue strength of the parts which are susceptible to hydrogen absorption during the plating cycle or associated acid/alkaline cleaning cycles. Induced hydrogen in the parts causes the hydrogen embrittlement phenomenon. Also, high residual tensile stresses produced during plating reduce fatigue strength of the base metal and the corrosion resistance of both the plated surface as well as the base metal. Pits formed due to improper cleaning also act as stress raisers for fatigue crack propagation.

Residual Stresses	Is potentially dangerous particularly in the critical service applications, Causes geometrical distortion, fatigue, and stress corrosion cracking.
Surface irregularities (geometric variation, local waviness, surface texture, toll marls, rough surface, etc.)	Causes stress concentration effect and adversely affects the fatigue strength.
Under cutting	Causes loss of dimension and a severe notch effect.
Etch pits/corrosion	Acts as notch for initiation of either corrosion fatigue or stress corrosion cracking.
Craters/rough surface	Are highly stress forming zones and adversely affect the fatigue strength.
Hard-chilled layer	Causes surface crazing, crack, etc. by virtue of extreme brittleness/ internal stresses and accelerate low cycle fatigue/ brittle failure.
crack	Reduces section thickness as well as strength bearing capacity affecting the fatigue resistance property drastically as a severe notch.
Heat affected zone/softening	Lowens the hardness value in the subsurface zone with loss of yield/tensile strength. Fatigue cracks usually originate in the transition zone between the heat affected zone and the hard white layer due to the existence of high tensile stresses.
Residual tensile stresses	Helps to diminish the fatigue resistance. Properly.
Improper welding	
Incomplete/inadequate fusion	Reduces the weld joint strength and fatigue strength.
Under bead cracks, hot or cold cracks	Affects fatigue strength and acts as a point of stress concentration.
Undercut	Reduces the weld thickness and strength properties. Also acts as a severe notch for development of fatigue crack.
Not homogenous microstructure (coarse and widmanstatten structure, oxide film, decarburized layer, carburized skin, alloy	Weakens the weld metal strength proper-ties, develops cracks (in the case of formation of martensite in the heat affected zone) and reduces fatigue strength property.

segregation, martensite in the heat affected zone, inclusion, gas porosity, etc.)	
Weld decay	Develops intergranular/stress corrosion cracking particularly in austenitic stainless steel.
Assembly Errors	
Operator's negligence	Results in costly damage interrupting seriously the progress of production.
Inadequate/inaccurate or ambiguous specification of procedure for assembly (non-uniform and improper tightening)	Introduces a relative movement between the parts and initiate fatigue cracking at the point of stress concentration due to fretting.
Misalignment/mismatch	Imposes additional load causing overstressing and overheating. Also initiates fatigue cracks. Stress-corrosion crack may result from undue straining due to mismatch in high strength aluminum alloys.
Improper Service Conditions	
Wear-adhesive and abrasive type	Causes dimensional deviation, introduces stress-concentration and susceptibility to fatigue.
Corrosion-oxidation, pitting and intergranular corrosion	Causes dimensional deviation and stress concentration and reduces strength and fatigue resistance.
Thermal-high temperature scaling, microstructural changes, grain growth, incipient melting	Reduces strength and thermal fatigue resistance and affects creep property.
Inadequate/bad maintenance and improper repair.	Results in premature failure or undetected defects during overhaul.